

Instructions:

- **Format:** Please submit a PDF file with typed answers (L^AT_EX or otherwise). Handwritten notes will not be graded.
- **Submission:** Only homework uploaded to Gradescope will be graded. Make sure to show all the steps of your derivations in order to receive full credit.
- **Integrity and Collaboration:** You are expected to work on the homework by yourself. You are not permitted to discuss them with anyone except the instructor. The homework that you hand in should be entirely your own work. You may be asked to demonstrate how you got any results that you report.
- **Clarifications:** If you have any questions, please look at Piazza first. Other students may have encountered the same problem, and is solved already. If not, post your question there. We will respond as soon as possible.

[20 points] Consider the following function:

$$g(x) = \begin{cases} 1 & \text{if } x \text{ is a power of 2} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

1. Show that the function $g(x)$ is not Riemann integrable on any interval $([a, b])$ where $(a < b)$.
2. Demonstrate that the function $g(x)$ is Lebesgue integrable and the result of the integral.

[20 points] Let $x_1, x_2, \dots, x_n$ be a set of d -dimensional input vectors.

1. Write an expression to find a vector $\hat{x}$ that minimizes the sum of l_2 -distance to all n input vectors. Show that $\hat{x}$ is equal to the mean value of the vectors.
2. Repeat part (a) using l_1 instead of l_2 distance. What should $\hat{x}$ correspond to?

[20 points] Using matrix calculus, compute the gradient of each function $f(\mathbf{x})$ or $f(\mathbf{X})$ below with respect to its argument $\mathbf{x}$ or $\mathbf{X}$. For full credit, you must show your derivation clearly, which includes expressing the matrix equation in terms of its vector or matrix components (elements) and the intermediate steps when computing the partial derivatives with respect to an element of the vector or matrix.

1. $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x} - \alpha \mathbf{z}^T \mathbf{x}.$

2. $f(\mathbf{X}) = \|\mathbf{A} - \mathbf{X} \mathbf{U} \mathbf{X}^T\|_F^2 + \lambda \|\mathbf{X}\|_F^2$

[20 points] Let $f(x, y, z) = x^2 y - 4x - y \sin(z)$ for $0 < z < \pi$. Find the critical points of this function. Classify the critical points as local maximum, local minimum, or saddle point. Show your justifications.